(a) both assertion and reason are true and reason is the correct explanation of assertion.

(b) both assertion and reason are true but reason is not the correct explanation of assertion.

(c) Assertion is true but reason is false

(d) Both assertion and reason are false

12. Heterolytic fission of C–C bond results in the formation of

(a) free radical (b) Carbanion

(c) Carbocation (d) Carbanion and Carbocation

13. Which of the following represent a set of nuclephiles?

(a) BF_3, H_2O, NH^{2-}

(b) $AlCl_3$, BF_3, NH_3

(c) CN^-, RCH_2^-, ROH

(d) H^+, RNH_3^+, $:CCl_2$

14. Which of the following species does not acts as a nucleophile?

(a) ROH (b) ROR

(c) PCl_3 (d) BF_3

15. The geometrical shape of carbocation is

(a) Linear (b) tetrahedral
(c) Planar (d) Pyramidal

II. Write brief answer to the following questions.

16. Write short notes on

(a) Resonance
(b) Hyperconjucation

17. What are electrophiles and nucleophiles? Give suitable examples for each.

18. Show the heterolysis of covalent bond by using curved arrow notation and complete the following equations. Identify the nucleophile is each case.

(i) CH_3 - Br + KOH $\rightarrow$

(ii) CH_3 - OCH_3 + HI $\rightarrow$

19. Explain inductive effect with suitable example.

20. Explain electromeric effect.

21. Give examples for the following types of organic reactions

(i) β - elimination

(ii) electrophilic substitution.

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